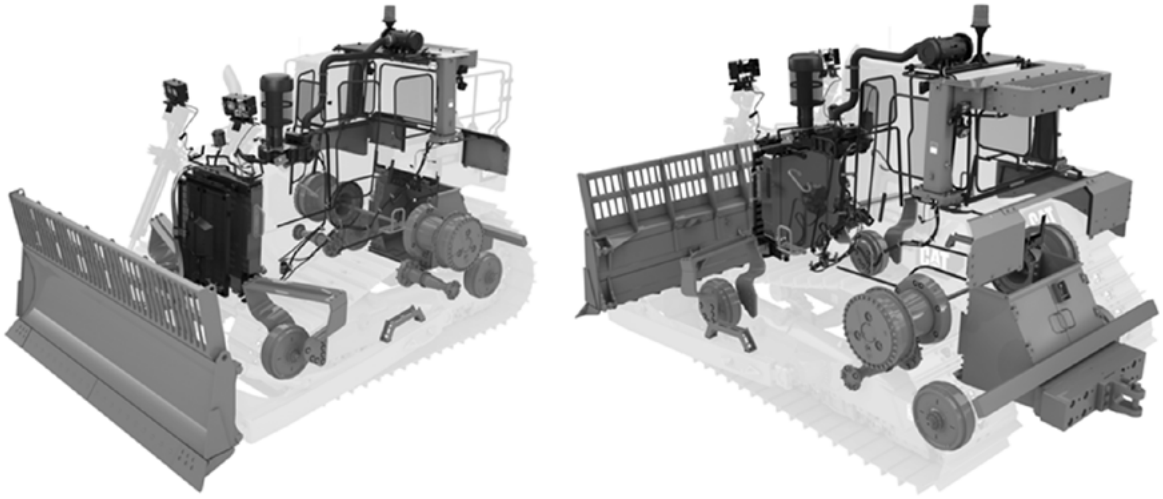
Cat Waste Handling Track-Type Tractor modifications are purpose built and installed at the original manufacturing facility prior to shipment.

Features:

- **Landfill blade** increases capacity
- **Front and rear striker bars** help protect machine from debris impact
- **Clamshell guards** on final drives and idler/final drive seals help prevent wire wrap and keep debris from wrapping around Duo-Cone™ seals.
- **Bottom guard, chassis and tilt cylinder guards,** and guarding over the fuel tank and battery box
- **Center-hole track shoes** help eject debris from the tracks.
- **Specialized air handling features** help deliver cleaner air to the machine and cab
- **Lights mounted up and away** from main debris area for protection, while providing plenty of light on the work area
- **Cooling system** designed for high debris environments and easy access for cleanout
- **Powered cab precleaner** is on back of cab to help increase cab pressurization and keep debris and odor out.
- **Debris sealing** in the machine improved
- **Waste cab** with dual pane windows, polycarb doors resistant to impact
- **Better door sealing** for improved cab pressurization
- **Integrated beacon lights**
- **Striker bar box** for Waste, no counterweight needed

WASTE GUARDING – PROTECT YOUR INVESTMENT



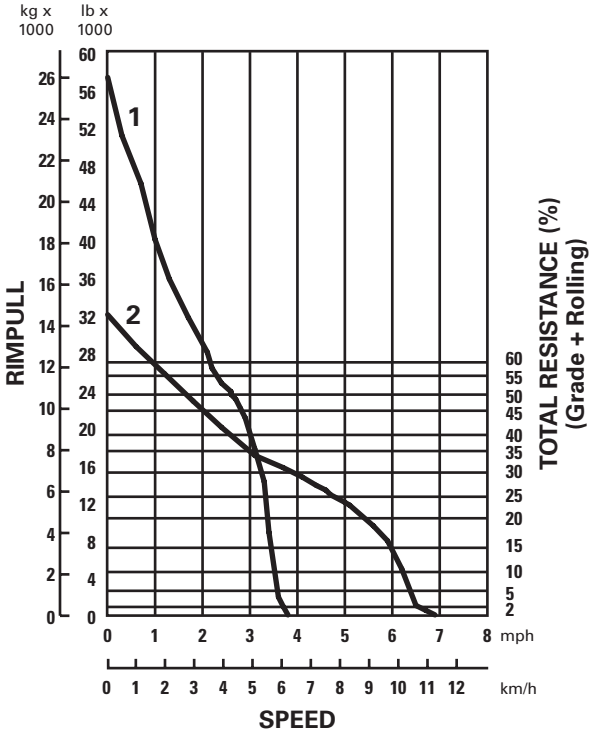
953, 963, 973 Waste Handlers

- **Waste Handler** is a versatile machine for loading, sorting, excavation and spreading cover, well suited to the landfill or transfer station.
- **Specialized guarding, striker bars and seals** help protect the machine and components from impact and airborne debris.
- **Final Drive guarding** helps prevent wrapping and damage.
- **Screen** helps protect windshield and operator from breakage and debris.
- **High debris cooling system** – radiator fan folds out for easy cleanout access.
- **Specialized air handling features** help deliver cleaner air to the machine and to the cab.
- **Choice of buckets and track shoes** allow you to further optimize the machine for greater performance and service life.
- **Enhanced Cleaning Package** reduces cleaning time, adds machine protection and increases compaction.
- **Track loader power, traction and agility** mean you can use one machine and one operator to do everything from clearing the site to installing utilities and finishing around buildings.
- **Fusion™ Quick Coupler** option lets you make fast attachment changes with a wide range of tools like forks, buckets, etc. from wheel loaders and other Fusion compatible machines.
- **Bucket/tooth, undercarriage, track shoe and attachment choices** help you equip your machine for optimal performance in your applications.
- **Purpose-built Waste Handler and Low Ground Pressure (LGP)** configurations stand up to the toughest tasks.

Features:

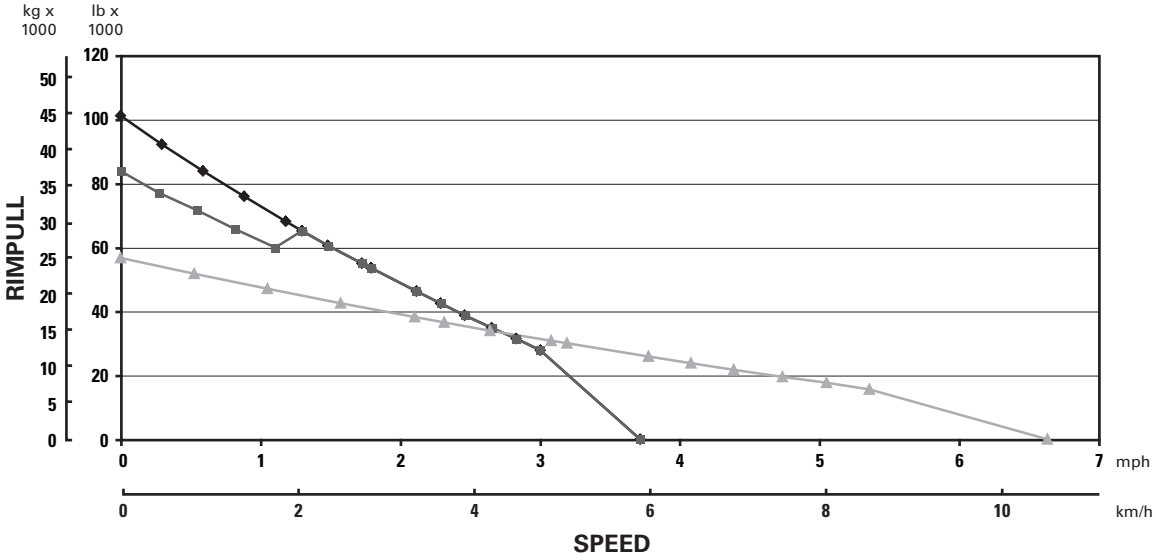
- **Caterpillar designed and manufactured power train** … for optimum match, performance and efficiency. Responsive Cat diesel engine. Single-lever planetary power shift. All-wheel drive.
- **Center-point articulation** … excellent maneuverability. Front and rear drums track, so material is chopped and compacted twice each pass.
- **Protective guarding** … helps keep trash from damaging machine components.
- **Cat landfill blades** spread refuse and cover material … built strong to handle the wide range of refuse encountered in landfills.
- **Operator comfort and convenience** … sound suppressed cab with pressurized and filtered air circulation system. Adjustable suspension seat. Electronic Monitoring System and gauge package is standard.
- **Striker bars** … standard on all Cat Landfill Compactors, prevents refuse from being carried over the rear wheels.
- **Caterpillar designed wheels** … We test and build a Cat system. Engineers who work together with our power train, structures and manufacturing engineers design and manufacture our wheels in the same facility in which the machines are designed and built. This ensures the entire system is complemented by each component. If you alter components, you could compromise a system that was designed and tested for peak performance. If a wheel is produced that does not meet our design specifications and does not balance the load over our final drives, it could reduce the life of the bearing substantially and wear out other components creating unnecessary downtime. This also allows our standard axle guard system to work with the components for which it was designed.
- **Wheel and tip configurations** … Three new wheel and tip configurations are available to meet your particular application:
  1) **Paddle tip** — High performance and less fuel burn with more traction and less weight.
  2) **Plus tip** — Traditional design for increased side slope stability and more mass for longer life.
  3) **Combination tip** — Both paddle and plus tips to give high performance with side slope stability.
  4) **Diamond tips** — Diamond tips are high performance and rated up to 15,000 hrs. These tips are the heaviest of all tips but designed to last longer.

816F2

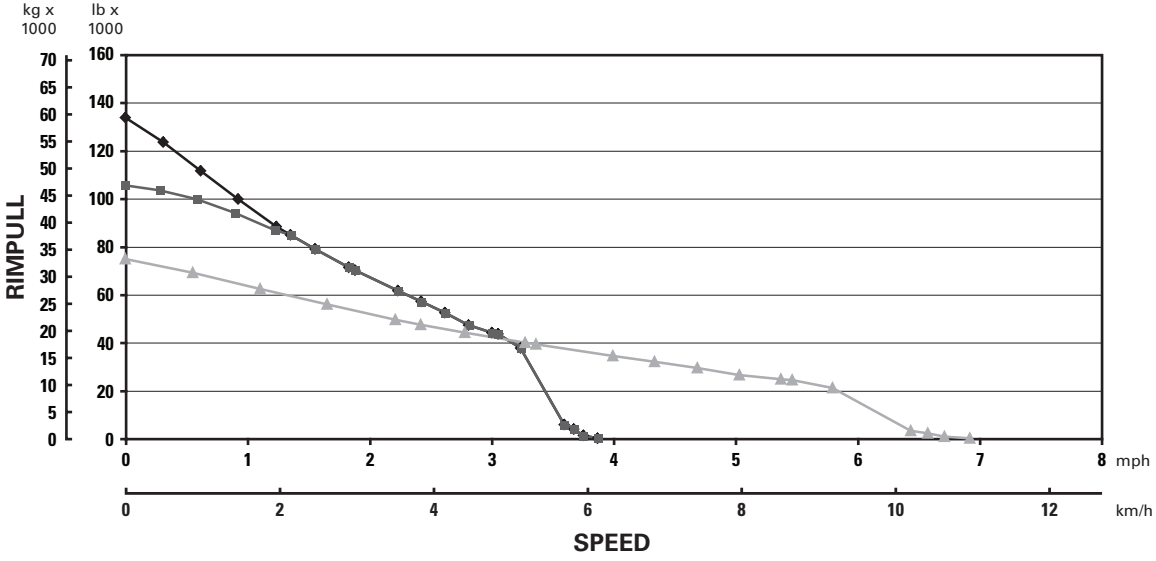


KEY
1 — 1st Gear
2 — 2nd Gear

826K



836K



KEY
- 1F CD
- 1F CD with PTS
- 2F CD

Features:

The section below is 950 – 980 MWL:

- **Tilt cylinder guard** – Heavy duty sliding guard mounts to the cylinder and protects the tilt cylinder rod from damage due to contact/impact from falling debris or material spillage over the work tool. If the rod(s) are pitted, scratched, the cylinder seals may be worn prematurely.
- **Axle seal guard** – Axle seal guard bolts to the axle shafts and protect each axle seal by preventing material, springs, wire, from cutting and damaging seals as material winds around when the tires rotate.
- **Front frame guard** – Guard bolts to the front frame to protect the caliper parking brake and prevent material accumulation front frame, protecting the front drive shaft.
- **Hinged powertrain guard** – Powertrain guard bolts underneath the rear frame to protect the transmission from contact with debris and prevents material ingress around the transmission. The guard is hinged to make servicing easier. An electric actuator available to provide easy removal of accumulated debris.
- **Hitch guard** – Hitch guard bolts to the butterfly plate to prevent debris from packing around the transmission and hydraulic pumps , while providing visibility to the transmission oil sight glass..
- **Steering cylinder guards** – Steering cylinder guards bolt to the frame just forward of the tires to protects the steering cylinders, sensors and pins from damage from debris from the rear tires.
- **Light guards (roading, work, rear)** – Light guards protect the front roading lights, cab work lights (standard halogen and premium lights) and rear stop/tail lights from damage by debris.
- **Steel rear deflectors** – The standard non-metallic rear deflectors are replaced with steel versions for use in extreme applications.
- **Narrow steel front fenders** – The narrow steel front fenders replace standard full-coverage fenders to provide more clearance to walls when the machine is operating in a confined place.
- **Reinforced service centers and platforms** – The electrical and hydraulic service centers are reinforced with additional bolted wear plates to resist damage to key components in the service centers. The service platforms are reinforced with welded-on plates to support additional weight for 3rd party suppression systems.
- **Heavy-Duty ladder and cable steps** – The heavy duty ladder replaces the standard ladder to provide additional resistance to damage when the machine is operated in a confined space. Steel cables replace the standard rubber strap and mates with the heavy duty steps to withstand the extreme nature of the application.
- **Turbine trash precleaner** – Cyclone engine air cleaner that uses centrifugal force to spin debris out of the air stream, extending the service interval for the air filter elements. A metallic screen prevents larger debris from clogging the filter.
- **Carbon fresh air filter** – Activated charcoal filter replaces the standard cab recirculation filter to reduce the odor causing debris particles in the operator environment.
- **Lift cylinder baffles** – Lift cylinder baffles prevent the accumulation of debris and compaction under the lift cylinders, protecting the cylinder from damage.

The section below is SWL features:

- **Windshield guard** – Windshield guard protects the cab glass from damage while working in tough waste applications. Guard can swing open for easy access to cleaning.
- **Tilt cylinder guards** – Heavy duty sliding tilt cylinder guards provide protection against airborne debris and potential bucket spill material from damaging the tilt cylinders.
- **Light guards** – Guards are provided to shield the front, rear and roading lights from debris damage while working in waste handling applications. Rear lights are not an option when rear waste gate is selected.
- **Narrow steel front fenders** – Standard full-coverage front fenders are replaced with a heavy duty narrow steel fender design. These robust fenders are designed specifically to live in a waste handling application.

- **Driveshaft guard** – Protecting the underside of the front half of the machine, these guards prevent debris build up and damage to major components such as the drive shaft.
- **Hitch guard** – Provides protection to hitch area of the rear frame, preventing debris from entering the frame while allowing visibility to gear box site gauge.

SMALL WHEEL LOADER GUARDING AND OPTIONAL EQUIPMENT 926-938 MACHINE MODELS

- **Steering cylinder guard** – Provides protection to both steering cylinders, preventing damage to rods and cylinder seals.
- **Power train side guard** – Mounts on side of machine to provide protection for the transmission components.
- **Power train lower guard** – Protecting the underside of the rear half of the machine, belly guards prevent debris build-up and damage to major components such as the engine and transmission. Side power train guard is an option as well.
- **Crankcase guard** – Mounts underneath machine to provide protection to the engine components.
- **Rear waste guard gate** – Heavy Duty radiator guard provides additional rear machine protection in this tough application. The guard is hinged for easy access to the airborne debris screen and cooling cores.
- **Reversing fan** – The reversing fan, with three settings, auto/off/manual along with a single plain, wide spaced and 100% sealed cooling package, will help clear out any debris that makes its way into the area. In the auto mode, the timing for the reversing fan can be customized to fit the application needs.
- **Sealed alternator** – A filtered, brushless alternator is ideal for dusty or corrosive environments/applications.
- **Turbine engine precleaner** – Turbine precleaner extends filter life and helps to provide clean air to the engine in high debris applications.
- **RESPA cab filtration** – RESPA cab filtration system is a powered cab precleaning system which supplies clean air to the cab in high debris application.

Recommended Waste Handling Options

- **Powered crankcase and power train guards** raise and lower at the flip of a switch allowing fast, easy and frequent cleanings.
- **High lift arrangement** increases bucket hinge pin height offering increased stacking and stockpiling capability. A critical feature when incoming volumes exceed conveyor capacity or floor space.
- **Traction control system (TCS)** option for 938H/IT38H provides maximum traction in slippery conditions. TCS electronically senses and limits wheel slip at each wheel independently.
- **Limited-slip differential** limits tire slip on both front and rear axles. Increases traction and reduces tire wear and scuffing in wet or dry conditions. Provides an alternative to the No SPIN differential which is not recommended due to increased tire wear, turning interference and poor tracking on dry surfaces.
- **Tire options:**
  L-5 bias hard rock lug
  L-5 slick
  RL-5K Goodyear radials
  XMINE Michelin radials (varies by model)
  Foam-filled
- **Hitch guards** help protect components in hitch area from damage.
- **Reversing radiator fan** is hydraulically driven and can be reversed by a switch in the operator station or automatically by timer. Reduces need for cooling system cleaning and improves cooling capability.
- **Other guarding and options** are available. Contact your Cat dealer for information.

Work Tools

- **Refuse bucket** has excellent dozing and stockpiling capability. Large spill plate helps protect the machine from debris falling over the top of the bucket. Available in pin-on or quick coupler configurations.
- **Pallet forks** are ideal for handling refuse destined for further recycling or stacking refuse in landfills for covering.
- **Multi-purpose bucket** has the capability to clamp and sort large objects, doze cover material or other light dozing chores.
- **Quick coupler** increases versatility by allowing a single machine to utilize a wide variety of work tools in a host of applications.

WORK TOOLS

As described in the opening statements of this section, the waste stream can consist of many different types of materials sometimes all blended together. For that reason, work tools become a very important part of moving, sorting, and dozing in waste applications. (Continue to consider the thought process of; "what is the application and material(s) you will work with — then, what tool(s) will accomplish your objectives, and finally, what machine(s) will utilize that tool to perform in that application?")

Wheel Loader Work Tools

MODEL	924K, 926M	930K, 930M	938K, 938M	950M, 962M, 950L, 962L, 950K, 962K, 950H, 962H	966M, 972M, 966L, 972L, 966K, 972K, 966H, 972H	980M, 980L, 980K, 980H
Bucket — Load & Carry	×	×	×	×	×	×
Bucket — Dozing	×	×	×	×	×	×
Bucket — Multi-Purpose	×	×	×	×	×	
Bucket — Waste Handling*	×	×	×	×	×	×

*The "Waste Handling" bucket is a current bucket that is being phased out and replaced by the Load & Carry and Dozing buckets. All are in the current price lists.

NOTE: Other attachments available upon request. Contact your Cat dealer.

NOTE: Learn more about available work tools @ Work Tool Central: *https://dealer.cat.com/products/wtc*.

Excavator Work Tools

MODEL	308E CR	315D L, 316E L	318E	319D L, 319D LN	M313D, M315D	M316D, M318D, M322D	M318D MH, M322D MH	320E L, 320E LRR, 321D CR, 323D	320C MH	M322D MH	324D, 324E
Grapple — Contractors'	×	×	×	×	×	×		×			×
Grapple — Demolition and Sorting		×	×	×	×	×		×			×
Grapple — Orange Peel							×		×	×	
Grapple — Trash		×	×	×	×	×		×			×

NOTE: Normal Waste Applications utilize the Trash, Sorting, and Orange Peel Grapples. Depending on the application, these work tools might be advisable.

NOTE: Learn more about available work tools in the *Hydromechanical Work Tools* chapter of the *Performance Handbook*.

NOTE: Learn more about available work tools @ Work Tool Central: *https://dealer.cat.com/products/wtc*.

(Chart continued on next page)

Excavator Work Tools (cont'd) MODEL	M325D MH, 320C MH	329D, 329E	336D, 336E	MH3049	345D, 349D, 349E	345C MH	MH3059	365C L	374D	385C	390D
Grapple — Contractors'		×	×		×			×	×	×	×
Grapple — Demolition and Sorting		×	×		×						
Grapple — Orange Peel	×			×		×	×				
Grapple — Trash		×	×		×						

NOTE: Normal Waste Applications utilize the Trash, Sorting, and Orange Peel Grapples. Depending on the application, these work tools might be advisable.

NOTE: Learn more about available work tools in the *Hydromechanical Work Tools* chapter of the *Performance Handbook*.

NOTE: Learn more about available work tools @ Work Tool Central: *https://dealer.cat.com/products/wtc*.

There are a variety of work tools for small–medium wheel loaders that have couplers. It is important to check the tasks that the wheel loader will perform and equip as needed. The 924–980 wheel loaders have three waste specific designed 'pin on' buckets to choose from.

- Load & Carry Bucket — An overall use bucket that is open and deep throated to gather and retain large loads. This bucket provides the ability to push, stockpile, load, and tamp no matter what type of loading is required.
- Doze Bucket — The height of this bucket provides more of a push platform to move heavy loads on the floor into waiting haul vehicles. Its main strengths are for pushing, stockpiling, and loading below grade vehicles.
- Tamp and Clamp Bucket — Designed with a longer snout and "optional" top clamps, this bucket is best for sorting, loading, and tamping haul vehicles with hard to handle materials. It has slightly less capacity than the load & carry and push buckets.

There are a variety of waste work tools for excavators and wheeled excavators. For Tracked and Wheeled Excavators you need to ask its primary use. Loading and Tamping, Sorting, Combination, Other Support?

- Trash Grapple — Designed to load and tamp large loads of material into haul vehicles. They can perform some material separation, but its main strength would be to load and tamp.
- Sorting Grapple — This grapple is designed to sort, separate, and load material. Equipped with a rotator, its main strength lies in the ability to separate material quickly, swing/rotate, and load haul vehicles, conveyors, and hoppers efficiently.
- Orange Peel Grapple — Also designed to sort, separate, and load material, its main strength is in a slightly larger load capability than the sorting grab. It also has the capability to rotate.
- Contractor's Grapple — Designed like the Waste Grapple, it is slightly smaller with a two over three tine configuration instead of four over five. It is heavier in construction and designed for materials heavier than MSW.

There is a wide variety of work tools for Excavators (wheeled and tracked) that can be used in waste applications; the above are "waste handling" tools designed for the application.

Support Work Tools

Brooms, pallet forks, bale clamps, blades, lifting arms, are just a few of the 'coupler attached' work tools that are used within the waste industry. A thorough assessment of primary and support roles of machine needs to be performed.

As stated in the chart, please contact your Dealer/Waste Representative or *Cat.com* websites for more information.

WHEEL DOZERS
SOIL COMPACTORS

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WHEEL DOZERS

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Tire Selection & Maintenance 19-4

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WHEEL DOZERS

Features:

- **Reliable Cat® power train:** four-stroke-cycle diesel with adjustment-free fuel system … full power shift with single lever on-the-go shifting.
- **Articulated frame steering** with hinge point midway between front and rear axles … short turning radius, long wheelbase … rear and front wheels track at all times.
- **Machine balance** … equal weight distribution on axles when blading.
- **All dozer functions,** including tip and tilt, hydraulically controlled from operator's seat.
- **STIC™ (Steering Transmission Integrated Control) Steering** is now offered on all Wheel Dozers.

Throughout this document, references to Tier 4 Interim/Stage IIIB/Japan 2011 (Tier 4 Interim) include U.S. EPA Tier 4 Interim, EU Stage IIIB, and Japan 2011 (Tier 4 Interim) equivalent emission standards. References to Tier 4 Final/Stage IV/Japan 2014 (Tier 4 Final) include U.S. EPA Tier 4 Final, EU Stage V, and Japan 2014 (Tier 4 Final) emission standards.

Throughout this document, references to Tier 1/Stage I include U.S. EPA Tier 1 and EU Stage I equivalent emission standards. References to Tier 2/Stage II/Japan 2001 (Tier 2) equivalent include U.S. EPA Tier 2, EU Stage II, and Japan 2001 (Tier 2) equivalent emission standards. References to Tier 3/Stage IIIA/Japan 2006 (Tier 3) equivalent include U.S. EPA Tier 3, EU Stage IIIA, and Japan 2006 (Tier 3) equivalent emission standards.

CONSIDERATIONS IN MACHINE SELECTION

The following factors should be considered when comparing wheels vs. tracks:

Traction

You can figure coefficient of traction, depending on underfoot conditions, from the Table Section in this book.

Wheels — up to 0.65 (in quarry pit with good floor)

Track — up to 0.90 (in soils permitting grouser penetration)

Usable Rimpull = Machine Weight × Coefficient of Traction

Speed

Wheels — travel speeds up to three times higher than track.

Maneuverability

Articulated steering and good visibility give wheel tractors high maneuverability.

Cost

See Owning and Operating Costs section. Tire vs. undercarriage costs can often be the deciding factor in selecting wheels or tracks.

Compaction

Ground Pressure:

Wheels — from 241 kPa (35 psi) to 310 kPa (45 psi)

Tracks — from 82 kPa (12 psi) to 97 kPa (14 psi)

Application

Utility … mobility, maneuverability and good speed suit wheel tractors for yard and stockpile work and for clean-up around shovels. Lower maintenance costs may be realized in certain soils that can be highly abrasive to track-type undercarriages.

Coal pile … recommend wheel tractors in this application when following conditions are present:

— Long push distances
— Need for good material spread
— High degree of compaction desired

Production Dozing … a wheel tractor should be considered in the following conditions:

— Long push distances
— Loose soils, little or no rock
— Level or downhill work
— Good underfoot conditions

Pushloading Scrapers … a wheel tractor should be considered in the following conditions:

— Thin scraper cut
— Good underfoot conditions — no rock
— Higher push speeds

Chip and Coal Scoops … a wheel dozer scoop should be considered in the following conditions:

— Long push distances
— Light, well processed materials such as coal or wood chips
— High degree of compaction desired
— Low to modest grades

COUNTERWEIGHTS AND BALLAST

For each specific application, there is a correct machine weight for proper balancing of traction, flotation, mobility and response.

- Low machine weight may increase tire slipping and wear, but improves flotation, mobility and machine response.
- High machine weight increases traction, but decreases mobility and response.

The machine weight is optimum for the operating conditions when wheel slipping barely occurs in the gear being used. Weight distribution under operating conditions should then be approximately equal between the wheels to balance power to each axle.

Application

Lower machine weight is usually required for applications such as fill spreading, stockpiling, road maintenance, towing compactors, and shovel cleanup. These are generally performed in either first or second gear. However, usage of second gear may involve a tradeoff of increased GET wear on the blade.

Higher machine weight is usually required for applications such as heavy dozing and push-loading, which are generally performed in first gear.

TIRE SELECTION & MAINTENANCE

Requirements of traction, flotation and tire life are met by a choice of tire size, tread design and inflation pressure.

Tire Width

For good conditions with little rolling resistance on surfaces where flotation is no problem, a narrower tire may be most economical. It may also be considered in muddy conditions in which the mud can be penetrated to reach firm earth underneath.

Where flotation problems and increased rolling resistance are encountered, wider tires are recommended. The greater contact area and shallower penetration increases flotation.

Tire Size

Larger optional tires will also improve flotation in soft conditions. With larger diameter, rimpull will be reduced which may be desirable to help control wheel spin.

Rock — Deep Tread (L-4) provides 50% more tread depth, thicker undertread and sidewall with increased tire life when compared to the L-3 tire. Recommended in rock conditions where sharp fragments cause high tire wear or sudden failures.

Rock — Extra Deep Tread (L-5) provides 150% more tread depth when compared to the L-3 tire. Intended for severe rock conditions with extreme penetration hazards.

Inflation Pressure

In average operating conditions the recommended inflation pressure prevents excessive deflection and minimizes tire rollover on side slopes.

Over-inflation

Reduces amount of tread contact with ground and provides less flotation. Over-inflation causes center of tread to wear faster and increases the chance of cuts and impact breaks.

Under-inflation

Can cause permanent tire damage in the form of flex breaks, radial cracks, and tread or ply separation. On jobs where wrinkling and bead rollover *are not* apparent, inflation pressure may be reduced to a minimum of:

Bias Ply	—	170 kPa (25 psi) on 35/65-33
		170 kPa (25 psi) on 29.5-25
		170 kPa (25 psi) on 26.5-25
		170 kPa (25 psi) on 23.5-25
Radial	—	310 kPa (45 psi) on 35/65-R33
		310 kPa (45 psi) on 29.5-R25
		205 kPa (30 psi) on 26.5-R25
		240 kPa (35 psi) on 23.5-R25

Reduced pressure will:

- Increase flotation and traction in sand.
- Improve envelopment characteristics to reduce sudden death failure on rock jobs.
- Provide better tread wear by reducing contact pressure between tire and ground.

Consult your tire manufacturer before changing tire pressures.

SOIL COMPACTORS

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Features:

- **Dozing, filling and compacting versatility.**
- **High speed operation** with responsive Cat diesel engine, single-lever planetary power shift transmission, and all-wheel drive.
- **Articulated frame** makes maneuvering quick and easy. Long wheel base for stability.
- **Wheels with tamping foot design and chevron pattern** give traction, penetration and compaction needed for high production. Foot pattern reversed on trailing drums to prevent overprinting lead drums.
- **Rear drums track front** for double compactive effort. Drum spacing covers mid-axle strip on return pass.
- **Rear axle oscillation** keeps all drums on ground for traction and stability.
- **Cleaner bars** keep drums free of carry over earth regardless of rolling direction. Adjustable, replaceable.
- **Optional fill spreading dozer** has single lever control for raise, lower, hold and float. (Blade tilt optional.)
- The 825K now offers **STIC (Steering Transmission Integrated Control) Steering** as a standard feature.

COMPACTION FUNDAMENTALS

The following discussion applies to soil compaction only.

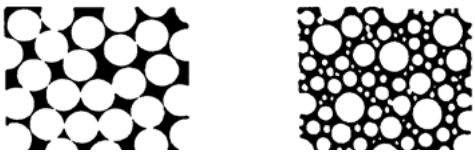
Definition

Compaction is the process of physically densifying or packing the soil … resulting in increased weight per unit volume. It is generally accepted that the strength of a soil can be increased by densification. Three important factors affect compaction.

— Material gradation
— Moisture content
— Compactive effort

Material Gradation — refers to the distribution (% by weight) of the different particle sizes within a given soil sample. A sample is *well-graded* if it contains a good, even distribution of particle sizes. A sample composed of predominantly one size particle, is said to be *poorly-graded*. In terms of compaction, a well-graded soil will compact more easily than one that is poorly-graded. In well-graded material the smaller particles tend to fill the empty spaces between the larger particles, leaving fewer voids after compaction.

MATERIAL GRADATION



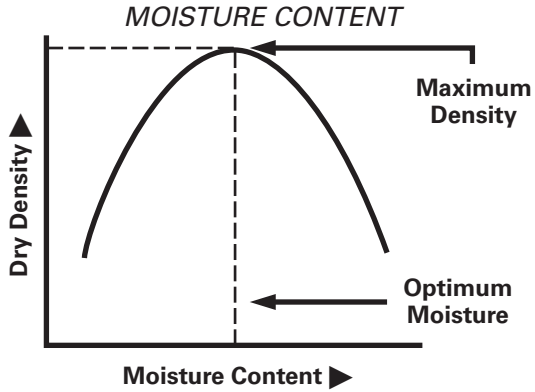
Poorly-graded Well-graded

Moisture Content — or the amount of water present in a soil, is very important to compaction. Water lubricates soil particles thus helping them slide into the most dense position. Water also creates clay particle bonding, giving cohesive materials their sticky qualities.

OPTIMUM MOISTURE

Heavy clay	17.5%
Silty clay	15.0%
Sandy clay	13.0%
Sand	10.0%
Gravel, sand, clay mix (pit run)	7.0%

Experience has shown that it is very difficult, if not impossible, to achieve proper compaction in materials that are too dry or too wet. Soil experts have determined that in practically every soil there is an amount of water, called optimum moisture content, at which it is possible to obtain maximum density with a given amount of compactive effort. The curve below shows this relationship between dry density and moisture content. It is called a compaction curve, moisture-density curve or Proctor curve.



Compactive Effort — refers to the method employed by a compactor to impart energy into the soil to achieve compaction. Compactors are designed to use one or a combination of the following types of compactive effort.

— Static weight (or pressure)
— Kneading action (or manipulation)
— Impact (or sharp blow)
— Vibration (or shaking)

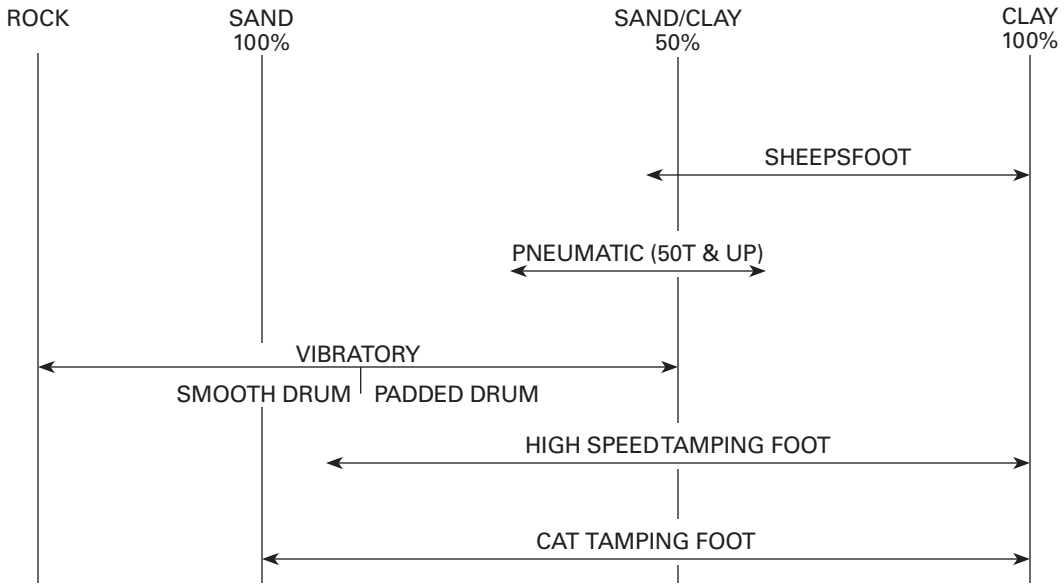
COMPACTOR TYPES

Compaction equipment can be grouped generally into the following classifications:
sheepsfoot
vibratory
pneumatic
high speed tamping foot
chopper wheels (see Landfill Compactor section)

Combinations of these types are also available, such as a vibrating smooth steel drum.

For ease of comparison, the compactors have been placed on the Zones of Application Chart shown below. This chart contains a range of material moistures from 100% clay to 100% sand, plus a rock zone. Each type has been positioned in what is considered to be its most effective and economical zone of application. However, it is not uncommon to find them working out of their zones. Exact positioning of the zones can vary with differing material conditions.

RANGES OF SOIL TYPES FOR SOIL COMPACTION EQUIPMENT



COMPACTOR PRODUCTION

Compactor production is expressed in compacted cubic meters (Cm^3) or compacted cubic yards (CCY) per hour. Material in its natural or bank state is measured in bank cubic meters or yards (Bm^3 or BCY). When it is removed or placed in a fill, it is measured in loose cubic meters or yards (Lm^3 or LCY).

When the loose material is worked into a compacted state, the relationship of *compacted material to bank material* is shown as the shrinkage factor (SF).

$$SF = \frac{\text{Compacted cubic meters } (Cm^3)}{\text{Bank cubic meters } (Bm^3)}$$

$$SF = \frac{\text{Compacted cubic yards (CCY)}}{\text{Bank cubic yards (BCY)}}$$

The construction industry has developed the following formula for use in estimating compactor production. This formula gives the material volume a given machine can compact in a 60-minute hour.

Metric Method

$$Cm^3 = \frac{W \times S \times L}{P}$$

W = Compacted width per pass, in meters. (For Cat Compactors it is recommended that W = Twice the width of one wheel.)

S = Average speed, in kilometers per hour.

L = Compacted thickness of lift, in millimeters.

P = Number of machine passes to achieve compaction **(can only be determined by testing the compacted material density on-the-job).**

English Method

$$CCY/hr = \frac{W \times S \times L \times 16.3}{P}$$

W = Compacted width per pass, in feet. (For Cat Compactors it is recommended that W = Twice the width of one wheel.)

S = Average speed, in miles per hour.

L = Compacted thickness of lift, in inches.

16.3 = Conversion constant, equals 5280 feet ÷ 12 inches ÷ 27 cubic feet

P = Number of machine passes to achieve compaction **(can only be determined by testing the compacted material density on-the-job).**

Example problem (Metric)

Determine production for an 815F2 operating under the following conditions:

P = 5, S = 10 km/h, L = 100 mm

Refer to 815K in the production table on the next page. Read down the first column until reaching section for 5 passes. Within this section in the second column, find the speed closest to 10 km/h. Read across this line to the 100 mm compacted lift. Read the production figure given.

Answer: 377 Cm^3/h. (Since the machine's speed of 10 km/h is slightly faster than the 9.5 of the table, production may be interpolated slightly higher — say 395 Cm^3/h.)

Example problem (English)

Determine production for an 825K operating under the following conditions:

P = 4, S = 8 mph, L = 6 inches

Refer to the production estimating table on the next page. This table contains estimates for the 815F2 and 825K Compactors using various speeds, lift thicknesses and number of passes. These figures were calculated using the formula discussed on this page. The figures represent 100% efficiency. W = Twice the width of one wheel.

In the 825K portion of this table, read down the first column until reaching the section for four passes. Within this section in the second column, find the line for 8 mph. Read across this line to the lift thickness column for 6 inches. Read the production figure given.

Answer: 1444 CCY/hr.

● ● ●

PRODUCTION TABLE

MODEL AND MACHINE PASSES*		AVERAGE SPEED		COMPACTED LIFT THICKNESS							
				100 mm	4 in	150 mm	6 in	200 mm	8 in	250 mm	10 in
		km/h	mph	m^3/h	yd^3/hr	m^3/h	yd^3/hr	m^3/h	yd^3/hr	m^3/h	yd^3/hr
815K	3	6.5	**4**	419	**548**	628	**822**	837	**1095**	—	
		9.5	**6**	628	**822**	942	**1232**	1256	**1643**	—	
		13.0	**8**	837	**1095**	1256	**1643**	1675	**2191**	—	
	4	6.5	**4**	314	**411**	471	**616**	628	**822**	—	
		9.5	**6**	471	**616**	706	**924**	942	**1232**	—	
		13.0	**8**	628	**822**	942	**1232**	1256	**1643**	—	
	5	6.5	**4**	251	**329**	377	**493**	502	**657**	—	
		9.5	**6**	377	**493**	565	**739**	754	**986**	—	
		13.0	**8**	502	**657**	754	**986**	1005	**1314**	—	
	6	6.5	**4**	286	**274**	314	**411**	419	**548**	—	
		9.5	**6**	314	**411**	471	**616**	628	**822**	—	
		13.0	**8**	419	**548**	628	**822**	837	**1095**	—	
825K	3	6.5	**4**	488	**642**	731	**962**	975	**1283**	1219	**1604**
		9.5	**6**	713	**962**	1069	**1444**	1425	**1925**	1781	**2406**
		13.0	**8**	975	**1283**	1463	**1925**	1950	**2566**	2438	**3208**
	4	6.5	**4**	366	**481**	534	**722**	731	**962**	914	**1203**
		9.5	**6**	534	**722**	802	**1083**	1069	**1444**	1336	**1804**
		13.0	**8**	731	**962**	1097	**1444**	1463	**1925**	1828	**2406**
	5	6.5	**4**	293	**385**	439	**577**	585	**770**	731	**962**
		9.5	**6**	428	**577**	641	**866**	855	**1155**	1069	**1444**
		13.0	**8**	585	**770**	878	**1155**	1170	**1540**	1463	**1925**
	6	6.5	**4**	244	**321**	366	**481**	488	**642**	609	**802**
		9.5	**6**	356	**481**	534	**722**	713	**962**	891	**1203**
		13.0	**8**	488	**642**	731	**962**	975	**1283**	1219	**1604**

*The number of machine passes required is dependent on soil type, moisture content, desired compaction and machine weight.

815K and 825K
Ground Contact Pressure/Soil Compactors

815K Tip	Weight Front Axle 9664 kg (21,305 lb) Ground Contact Pressure		Weight Rear Axle 12 722 kg (28,047 lb) Ground Contact Pressure		Contact Area Four Wheels	
Tip Penetration	**kPa**	**psi**	**kPa**	**psi**	**cm²**	**in²**
12.5 mm **(0.5 in)**	4727.05	**685.6**	6989.35	**706.7**	425.81	**66**
25 mm **(1.0 in)**	1347.92	**195.5**	1827.94	**215.7**	1445.16	**224**
38 mm **(1.5 in)**	902.52	**130.9**	1094.20	**156.8**	2077.42	**322**
50 mm **(2.0 in)**	658.45	**95.5**	872.95	**97.7**	3064.51	**475**

825K Standard Tip	Weight Front Axle 14 469 kg (31,899 lb) Ground Contact Pressure		Weight Rear Axle 21 059 kg (46,427 lb) Ground Contact Pressure		Contact Area Four Wheels	
Tip Penetration	**kPa**	**psi**	**kPa**	**psi**	**cm²**	**in²**
12.7 mm **(0.5 in)**	7178.41	**1041.14**	8092.55	**1173.73**	407.65	**63.19**
25 mm **(1.0 in)**	2609.39	**378.46**	2941.72	**426.66**	1121.55	**173.84**
38 mm **(1.5 in)**	1411.35	**204.70**	1591.10	**230.77**	2073.54	**321.40**
50 mm **(2.0 in)**	704.99	**102.25**	794.76	**115.27**	4150.96	**643.40**
75 mm **(3.0 in)**	610.19	**88.50**	687.89	**99.77**	4795.60	**743.32**
100 mm **(4.0 in)**	421.68	**61.16**	475.39	**68.95**	6939.86	**1075.68**
125 mm **(5.0 in)**	382.52	**55.48**	431.27	**62.55**	7650.04	**1185.76**
150 mm **(6.0 in)**	324.33	**47.04**	365.63	**53.03**	9022.18	**1398.44**
175 mm **(7.0 in)**	311.09	**45.12**	350.74	**50.87**	9405.66	**1457.88**
200 mm **(8.0 in)**	139.55	**20.24**	157.34	**22.82**	20 965.89	**3249.72**

825K Heavy Duty Tip	Weight Front Axle 14 469 kg (31,899 lb) Ground Contact Pressure		Weight Rear Axle 21 059 kg (46,427 lb) Ground Contact Pressure		Contact Area Four Wheels	
Tip Penetration	**kPa**	**psi**	**kPa**	**psi**	**cm²**	**in²**
12.7 mm **(0.5 in)**	7615.41	**1104.52**	8585.20	**1245.18**	96.07	**14.89**
25 mm **(1.0 in)**	6199.83	**899.21**	6989.35	**1013.72**	472.00	**73.16**
38 mm **(1.5 in)**	3614.20	**524.19**	1915.27	**277.79**	430.61	**66.74**
50 mm **(2.0 in)**	1621.44	**235.17**	1827.94	**265.12**	1804.64	**279.72**
75 mm **(3.0 in)**	970.64	**140.78**	1094.20	**158.70**	3014.96	**467.32**
100 mm **(4.0 in)**	774.28	**112.30**	872.95	**126.61**	3779.35	**585.80**
125 mm **(5.0 in)**	570.89	**82.80**	643.56	**93.34**	5126.18	**794.56**
150 mm **(6.0 in)**	443.13	**64.27**	499.59	**72.46**	6603.60	**1023.56**
175 mm **(7.0 in)**	417.06	**60.49**	470.22	**68.20**	7016.24	**1087.52**
200 mm **(8.0 in)**	389.07	**56.43**	438.64	**63.62**	7520.76	**1165.72**
225 mm **(9.0 in)**	381.07	**55.27**	429.61	**62.31**	7678.95	**1190.24**
250 mm **(10.0 in)**	128.59	**18.65**	145.00	**21.03**	22 753.76	**3526.84**

WHEEL LOADERS

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SPECIFICATION DEFINITIONS FOR FRONT END LOADERS

Cat wheel and track loader specifications conform to Society of Automotive Engineers (SAE) definitions as expressed in standards J732 (JUN92), as follows:

Description of Specification Machine

On wheel loaders the tire inflation pressure at which specifications are taken must be described in addition to the current written basic machine description. On track loaders the type of grouser must be specified.

Hydraulic Cycle Times

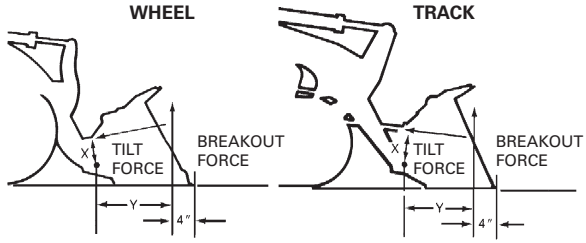
a. "Raise Time" — Time in seconds required to raise the bucket from level position on the ground.
b. "Lower Time" — Time in seconds required to lower the empty bucket from the full height to a level position on the ground.
c. "Dump Time" — Time in seconds required to move the bucket at maximum height from the maximum rollback position to full dump position while dumping the SAE loose material operating load.

Breakout Force

"Breakout force," pounds (and kilonewtons or kilograms) — the maximum sustained vertical upward force exerted 100 mm (4") behind the tip of the bucket cutting edge and achieved through the ability to lift and/or rollback about the specified pivot point under the following conditions:

a. Loader on a hard level surface with transmission in neutral.
b. All brakes released.
c. Unit at standard operating weight — rear of loader not tied down.
d. Bottom of cutting edge parallel to and not more than 20 mm (0.75") above or below the ground line.
e. When bucket circuit is used the pivot point must be specified as the bucket hinge pin, and the unit blocked under the bucket hinge pin pivot point in order to minimize linkage movement.
f. When the lift circuit is used, the pivot point must be specified as the lift arm hinge pin. Wheel loaders shall have front axle blocked to eliminate change in position of pivot pins due to tire deflection.
g. If both circuits are used simultaneously, the dominating pivot point listed in (e) or (f) must be specified.
h. If the circuit used causes the rear of the vehicle to leave the ground, then the vertical force value required to raise the rear of the vehicle is the breakout force.
i. For irregular shaped buckets, the tip of the bucket cutting edge referred to above shall mean the farther forward point of the cutting edge.

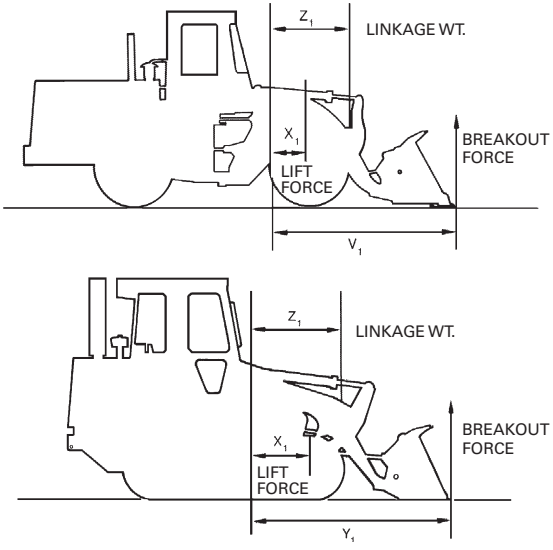
The following are illustrations used (according to provisions of SAE J732 JUN92) to measure Cat Loader breakout forces.



a. Breakout force resulting from rack back:
(Tilt Force) × (Dist. "X") = ("Y" Dist.) × (Breakout Force)

$$\frac{\text{(Tilt Force)} \times \text{(Dist. "X")}}{\text{"Y" Dist.}} = \text{Breakout Force}$$

b. Breakout force resulting from bucket lift:



(Lift Force) × (Dist. "X_1") = ("Y_1" Dist.) × (Breakout Force) + (Linkage Wt.) × (Dist. "Z_1") + (Breakout Force) × (Linkage Mechanical Advantage) "V_1"

or

$$\text{Breakout Force} = \frac{\text{(Lift Force)} \times \text{(Dist. "}X_1\text{")} - \text{(Linkage Wt.)} \times \text{(Dist. "}Z_1\text{")}}{\text{(Dist. "}Y_1\text{")} + \text{(Dist. "}V_1\text{")} \times \text{(Linkage Mech. Advantage)}}$$

Static Tipping Load

The minimum weight at center of gravity of "SAE Rated" load in bucket which will rotate rear of machine to a point where, on track loaders, front rollers are clear of the track and on wheel loaders, rear wheels are clear of the ground under the following conditions:

a. Loader on hard level surface and stationary.
b. Unit at standard operating weight.
c. Bucket at maximum rollback position.
d. Load at maximum forward position during raising cycle.
e. For articulated wheel loaders, the test will be run both with frame straight (straight static tipping load) and fully turned to a specific angle (full turn static tipping load).
f. Unit with standard equipment as described in specifications unless otherwise noted under the heading.

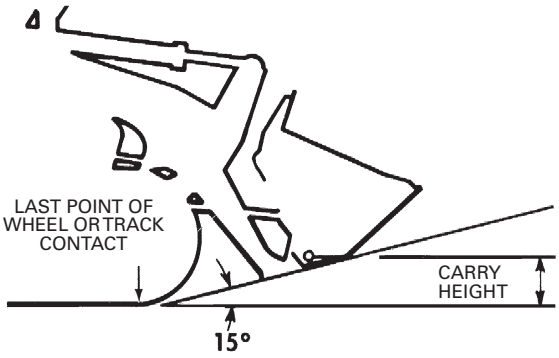
Operating Load

In order to comply with SAE standard J818 MAY87, the operating load of Wheel Loaders should not exceed 50% of the full turn Static Tipping load of the machine when equipped with attachments needed for the job. (For track loaders, operating load should not exceed 35% of the Static Tipping load rating.) See "Performance Data" of each machine in this handbook for increases to static tipping load by adding cab, counterweights, ripper-scarifier, etc.

The SAE operating load is not an indication of a wheel loader's rated payload. It takes into consideration only hydraulic lift and tipping capacity. There is no regard to structural and/or component lives, and for wheel loaders is measured on hard, moderately smooth and level operating surfaces.

Carry Position

SAE defines carry positions as: "The vertical distance from the ground to the center line of the bucket hinge pin, with the angle of approach at 15°." The sketch below illustrates this definition:



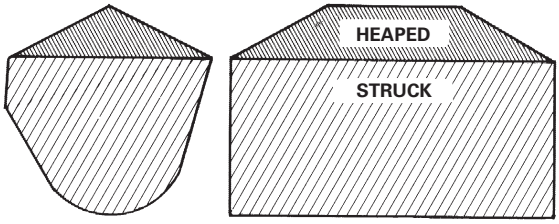
Loader Clearance Circle

SAE J732 JUN92 states that "minimum turning radius (over tire)" and "loader clearance circle" should be given for wheel loaders. Both are given on Cat specification sheets, including loader clearance circles for all available buckets for each machine.

Digging Depth

J732 JUN92 specifies digging depth as "the vertical distance in mm (inches) from the ground line to the bottom of the bucket cutting edge at the lowest position with the bucket cutting edge horizontal."

SAE BUCKET RATING



SAE Bucket Capacities

Struck capacity is that volume contained in a bucket after a load is leveled by drawing a straight edge resting on the cutting edge and the back of the bucket.

Heaped capacity is a struck capacity plus that additional material that would heap on the struck load at a 2:1 angle of repose with the struck line parallel to the ground.

SAE J742 (FEB85) specifies that the addition of any auxiliary spill guard to protect against spillage which might injure the operator will not be included in bucket capacity ratings. Buckets with irregular shaped cutting edges (vee edge) the strike plane should be drawn at one-third the distance of the protruding portion of the cutting edge. Cat rock buckets are built with integral see-through rock guards. Cat light material buckets come standard with bolt-on edges. These features which add to actual bucket capacity are included in published ratings.

Dump Height

SAE J732 JUN92 specifies that dump height is the vertical distance from the ground to the lowest point of the cutting edge with the bucket hinge pin at maximum height and the bucket at a 45° dump angle. Dump angle is the angle in degrees that the longest flat section of the inside bottom of the bucket will rotate below horizontal.

SELECTING A MACHINE

Steps in selecting the proper size loader:

1. Determine production required or desired.
2. Determine loader cycle time and cycles per hour. A machine size must be assumed to select a basic cycle time.
3. Determine required payload per cycle in loose cubic yards and pounds (meters and kilograms).
4. Determine bucket size needed.
5. Make machine selection using bucket size and payload as criteria to meet production requirements.
6. Compare the loader cycle time used in calculations to the cycle time of the machine selected. If there is a difference, rework the process beginning at step 2.

1. Production Required

The production required of a wheel or track loader should be slightly greater than the production capability of the other critical units in the earth or material moving system. For example, if a hopper can handle 300 tons per hour, a loader capable of slightly more than 300 tons should be used. Required production should be carefully calculated so the proper machine and bucket selections are made.

2. Loader Cycle Times

When hauling loose granular material on a hard smooth operating surface, a .45-.55 minute basic cycle time is considered reasonable for Cat articulated loaders with a competent operator. This includes load, dump, four reversals of direction, full cycle of hydraulics and minimum travel.

Material type, pile height, and other factors may improve or reduce production, and should be added to or subtracted from the basic cycle time when applicable.

When hauls are involved, obtain the haul and return portion of the cycle from the estimated travel chart (this section). Add the haul and return times to the estimated basic cycle time to obtain total cycle time.

CYCLE TIME FACTORS

A basic cycle time (Load, Dump, Maneuver) of .45-.55 minutes is average for an articulated loader [the basic cycle for large loaders, 3 m^3 (4 yd^3) and up, can be slightly longer], but variations can be anticipated in the field. The following values for many variable elements are based on normal operations. Adding or subtracting any of the variable times will give the total basic cycle time.

	Minutes added (+) or Subtracted (–) From Basic Cycle
Machine	
— Material handler	–.05
Materials	
— Mixed	+.02
— Up to 3 mm (1/8 in)	+.02
— 3 mm (1/8 in) to 20 mm (3/4 in)	–.02
— 20 mm (3/4 in) to 150 mm (6 in)	.00
— 150 mm (6 in) and over	+.03 and Up
— Bank or broken	+.04 and Up
Pile	
— Conveyor or Dozer piled 3 m (10 ft) and up	.00
— Conveyor or Dozer piled 3 m (10 ft) or less	+.01
— Dumped by truck	+.02
Miscellaneous	
— Common ownership of trucks and loaders	Up to –.04
— Independently owned trucks	Up to +.04
— Constant operation	Up to –.04
— Inconsistent operation	Up to +.04
— Small target	Up to +.04
— Fragile target	Up to +.05

Using actual job conditions and the above factors, total cycle time can be estimated. Convert total cycle time to cycles per hour.

$$\textit{Cycles per hour at 100\% Efficiency} = \frac{60\ \textit{min}}{\textit{Total Cycle Time in Minutes}}$$

Job efficiency is an important factor in machine selection. Efficiency is the actual number of minutes worked during an hour. Job efficiency accounts for bathroom breaks and other work interruptions.

$$\text{Cycles per hour at 50 minutes per hour (83\% efficiency)} = \text{Cycles per hour at 100\% efficiency} \times \frac{\text{50 min actual work time}}{\text{60 min hour}}$$

TRUCK LOADING

Average loader cycle times

910-962	0.45-0.50 min
966-980	0.50-0.55 min
986-990	0.55-0.60 min
992-994	0.60-0.70 min

3. Required Payload Per Cycle

Required payload per cycle is determined by dividing required hourly production by the number of cycles per hour.

4. Bucket Selection

After required payload per cycle has been calculated, the payload should be divided by the loose cubic yard (meter) material weight to determine number of loose cubic yards (meters) required per cycle.

The bulk of material handled does not weigh 1800 kg/m^3 (3000 lb/yd^3), so a reasonable knowledge of material weight is necessary for accurate production estimates. The Tables Section has average weight for certain materials when actual weights are not known.

The percentage of rated capacity a bucket carries in various materials is estimated below. The bucket size required to handle the required volume per cycle is found with the aid of the percentage of rated bucket capacity called "Bucket Fill Factor."

The bucket size needed is determined by dividing loose cubic meters (or yards) required per cycle by the bucket fill factor.

$$\textit{Bucket size} = \frac{\textit{Volume Required/Cycle}}{\textit{Bucket Fill Factor}}$$

BUCKET FILL FACTORS

The following indicates the approximate amounts of material as a percent of rated bucket capacity which will actually be delivered per bucket per cycle. This is known as "Bucket Fill Factor."

Loose Material	Fill factor
Mixed moist aggregates	95-100%
Uniform aggregates up to 3 mm (1/8 in)	95-100
3 mm (1/8 in) to 9 mm (3/8 in)	90-95
12 mm (1/2 in) to 20 mm (3/4 in)	85-90
24 mm (1.0 in) and over	85-90